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**Profaca**

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(54) **PRINTING UNIT WITH COMPACT CABLE MANAGEMENT**

(71) Applicant: **MEMJET TECHNOLOGY LIMITED**, Leopardstown (IE)

(72) Inventor: **Mark Profaca**, North Ryde (AU)

(73) Assignee: **Memjet Technology Limited**, Dublin (IE)

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(51) **Int. Cl.**

| | |
|---|---|
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| ***B41J 2/175*** | (2006.01) |
| ***B41J 25/308*** | (2006.01) |

(52) **U.S. Cl.**

CPC .......... ***B41J 29/02*** (2013.01); ***B41J 2/17523*** (2013.01); ***B41J 25/308*** (2013.01); ***B41J 29/023*** (2013.01); *B41J 2/175* (2013.01)

(58) **Field of Classification Search**

CPC ...... B41J 29/02; B41J 2/17523; B41J 25/308; B41J 29/023; B41J 2/175

See application file for complete search history.

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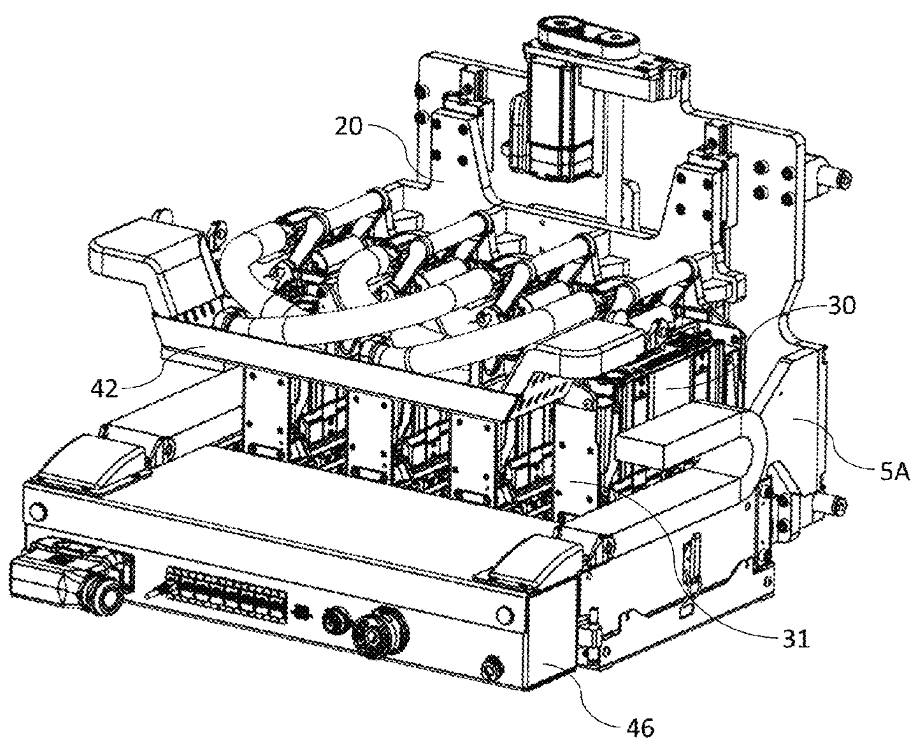
*Primary Examiner* — Justin Seo

(74) *Attorney, Agent, or Firm* — Garson & Gutierrez, PC

(57) **ABSTRACT**

A printing unit includes: a fixed unit chassis having a lower cable tray mounted to a front part thereof; a movable printhead chassis having an upper cable tray mounted to a front face thereof, the printhead chassis supporting an upstream print module and a downstream print module relative to a media feed direction; a lift mechanism for lifting and lowering the printhead chassis relative to the unit chassis; upstream and downstream cable trunkings positioned at opposite upstream and downstream ends of the unit chassis for receiving electrical cabling from the lower cable tray; and upstream and downstream conduits connected to the upstream and downstream print modules respectively. The downstream cable trunking feeds cabling from the lower cable tray into the upstream conduit via the upper cable tray, and the upstream cable trunking feeds cabling from the lower cable tray into the downstream conduit via the upper cable tray.

**6 Claims, 6 Drawing Sheets**



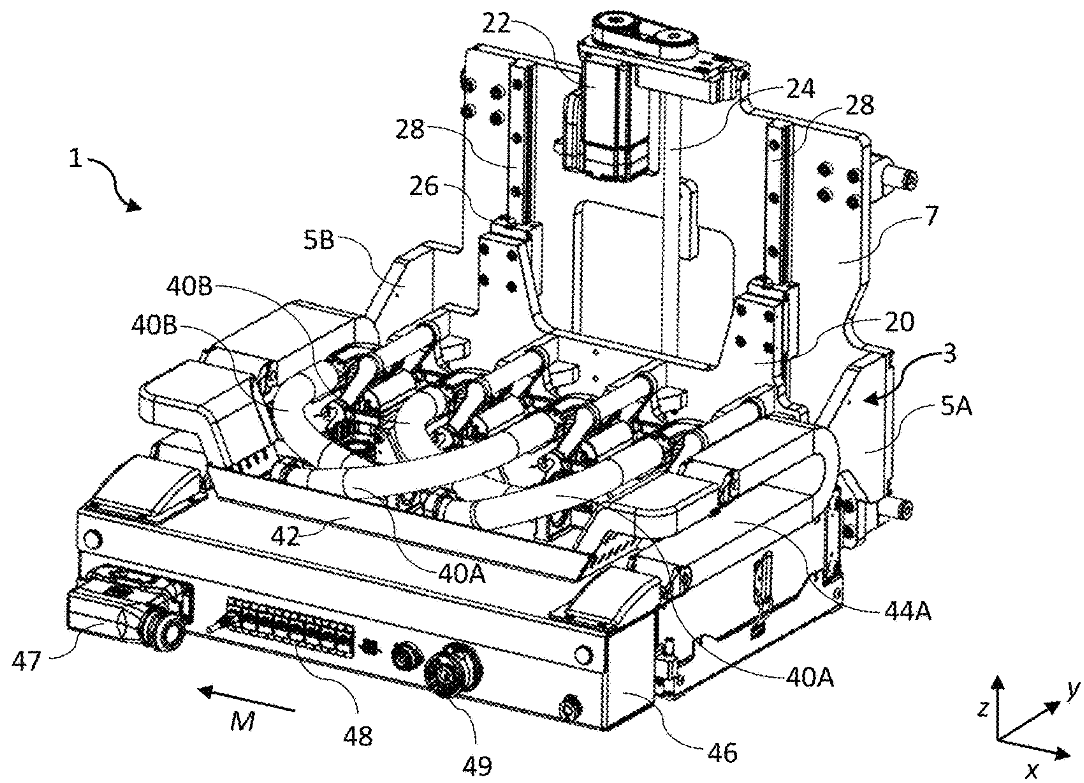

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FIG. 1

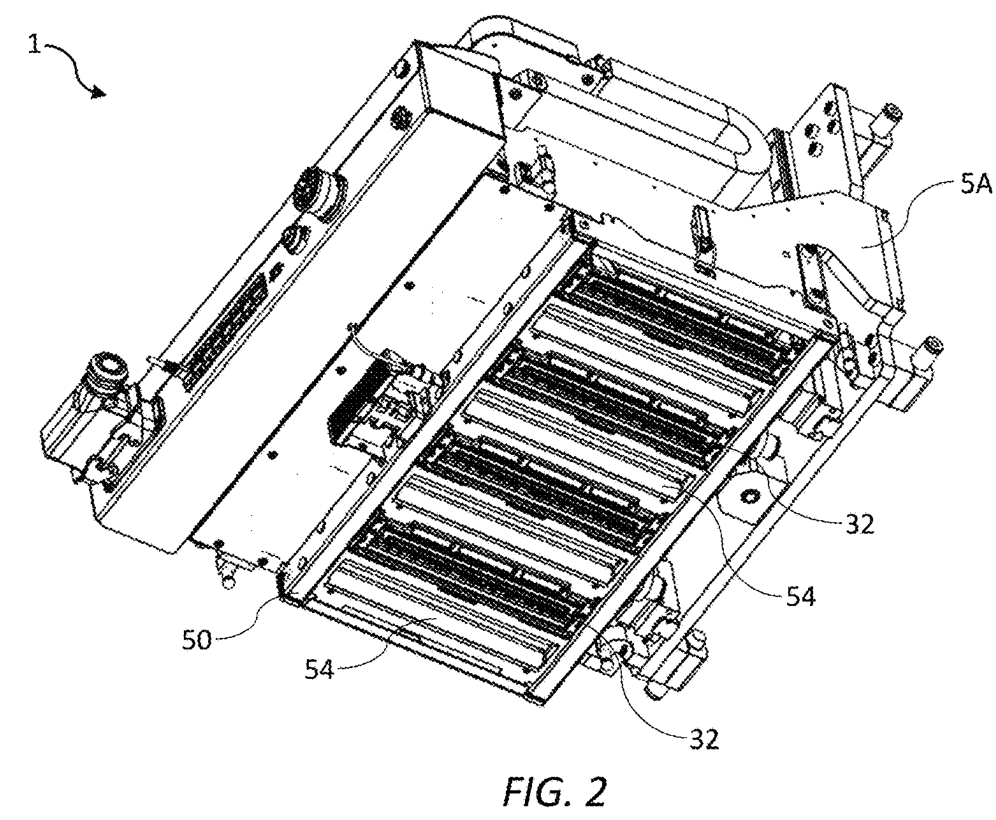
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FIG. 2

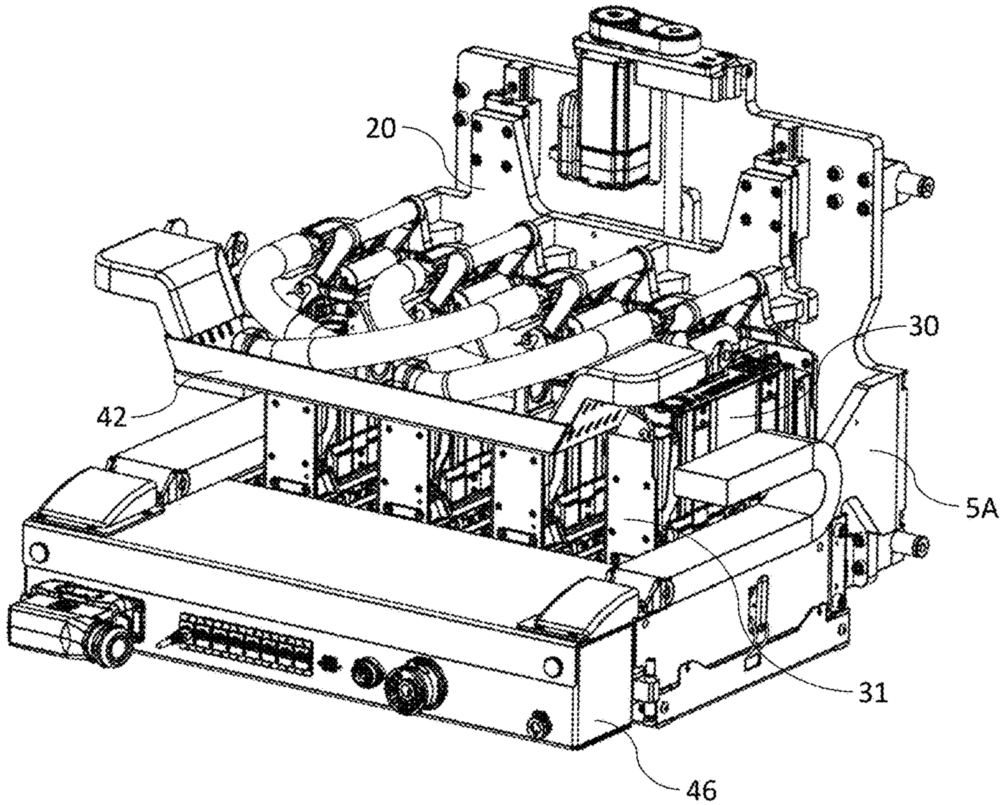


FIG. 3

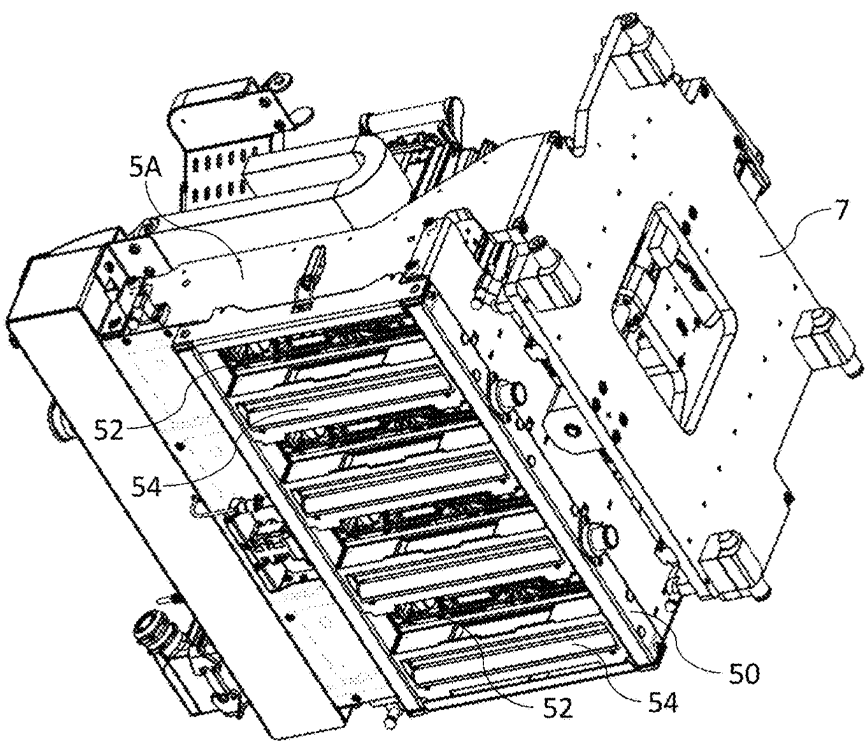


FIG. 4

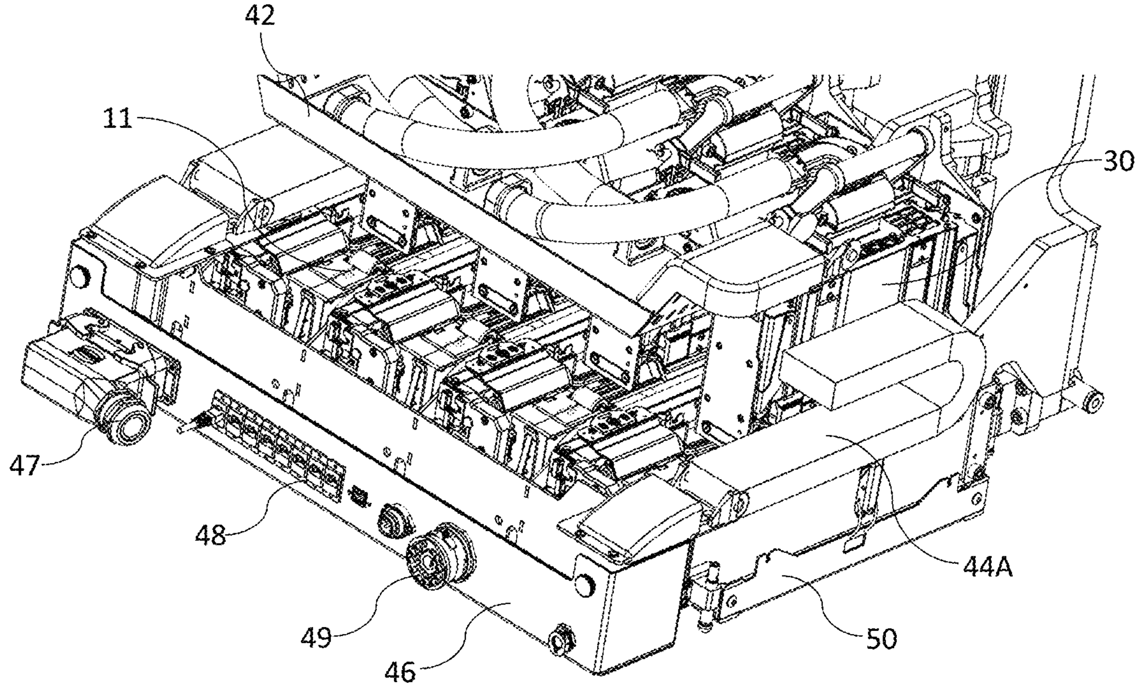


FIG. 5

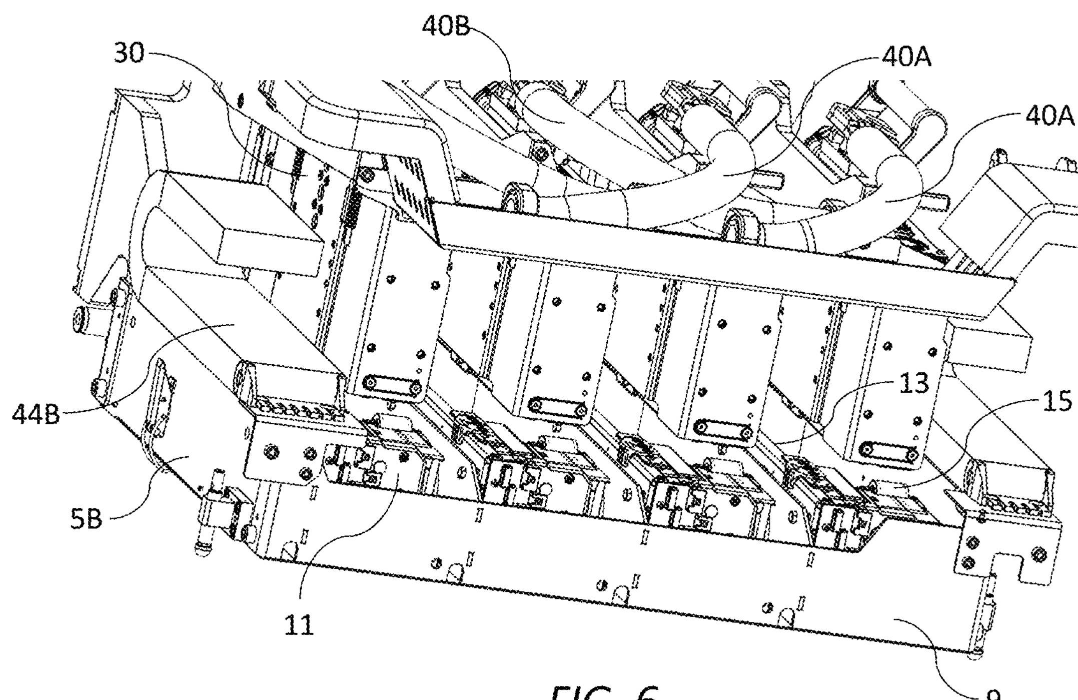


FIG. 6

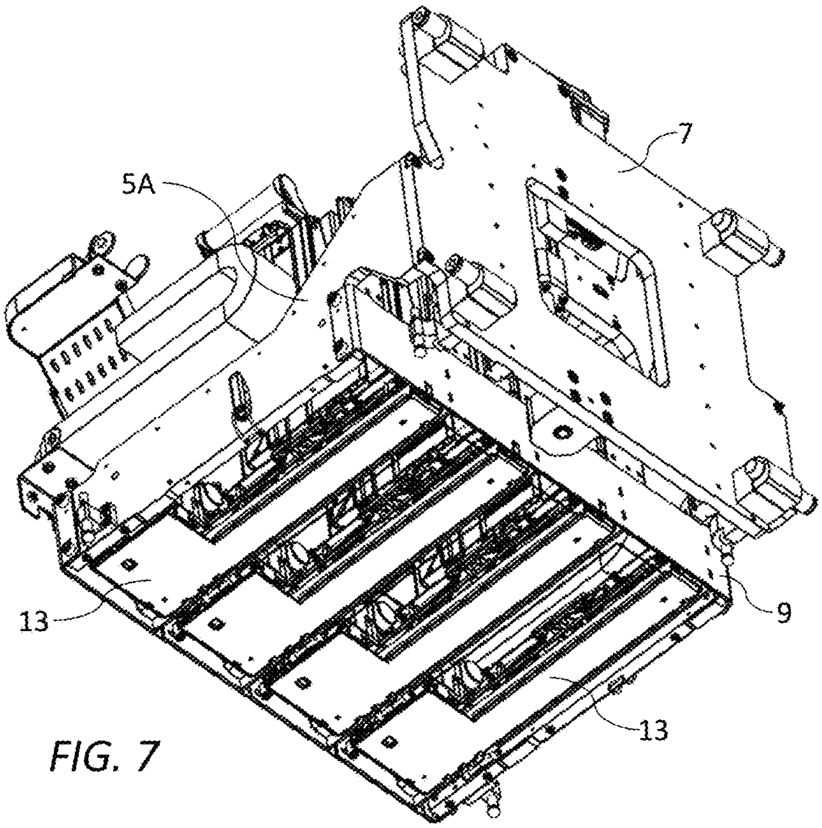


FIG. 7

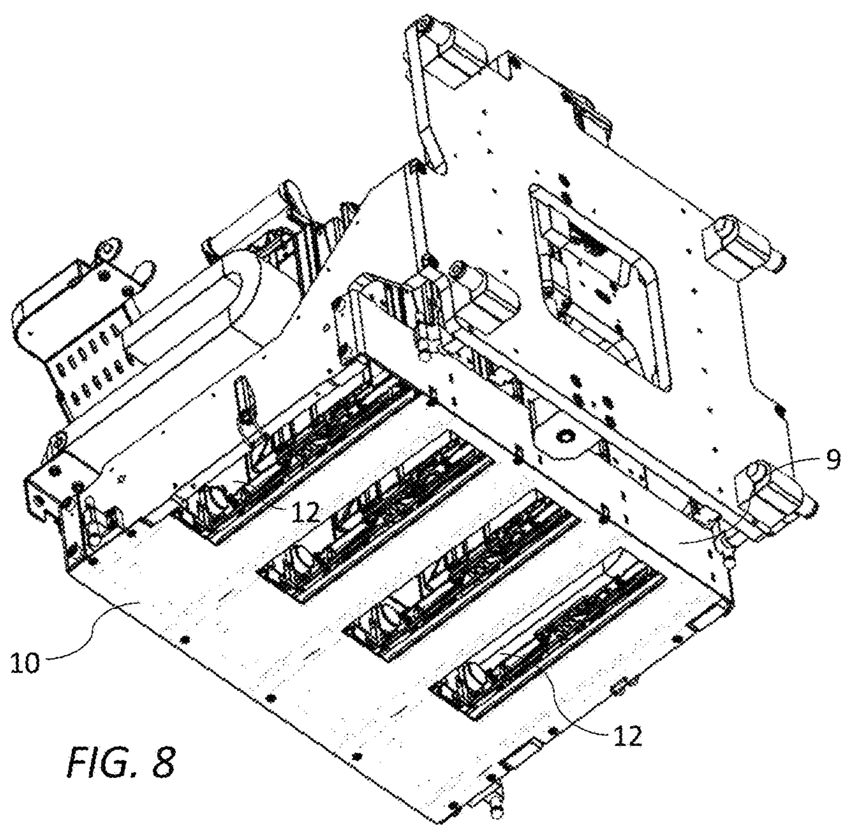


FIG. 8

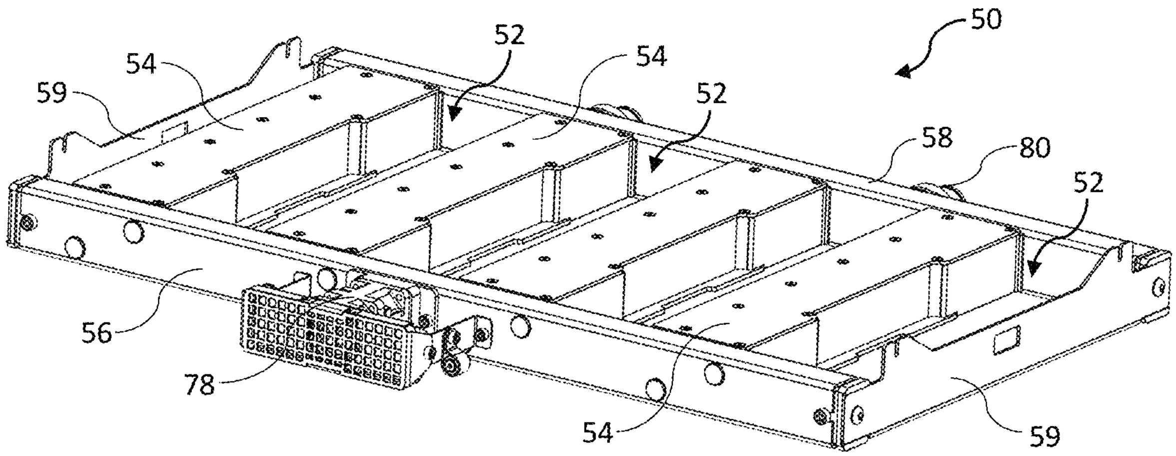
50
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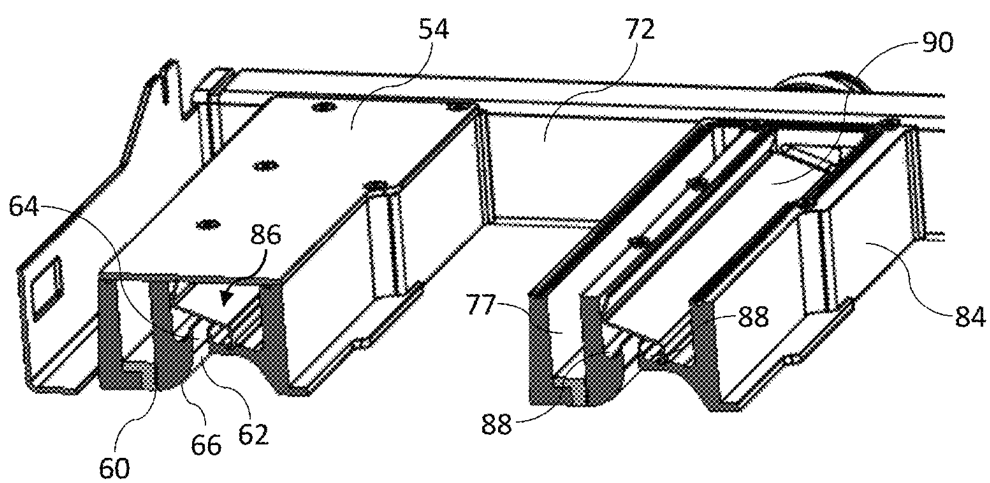
FIG. 9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60
66
62

FIG. 10

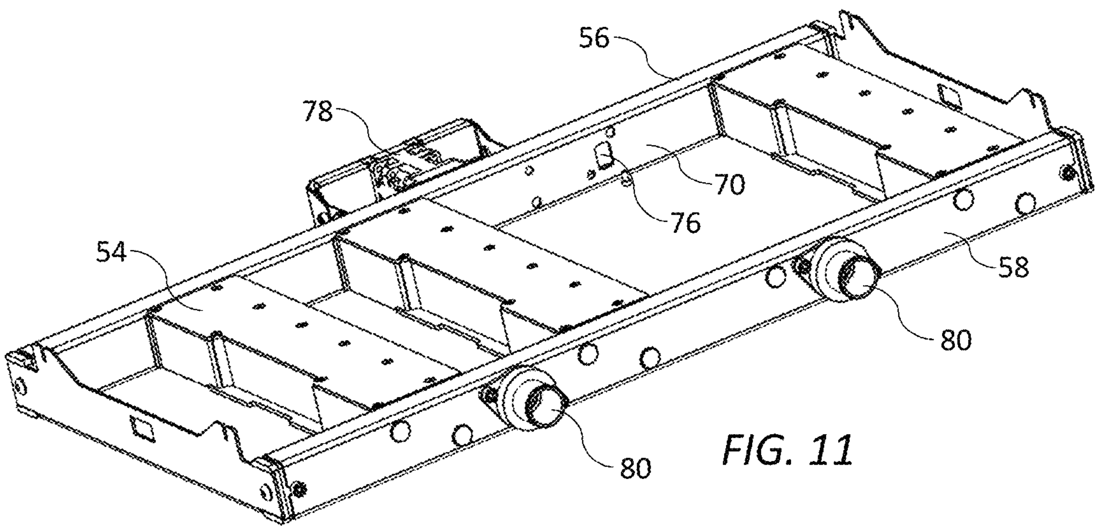


FIG. 11

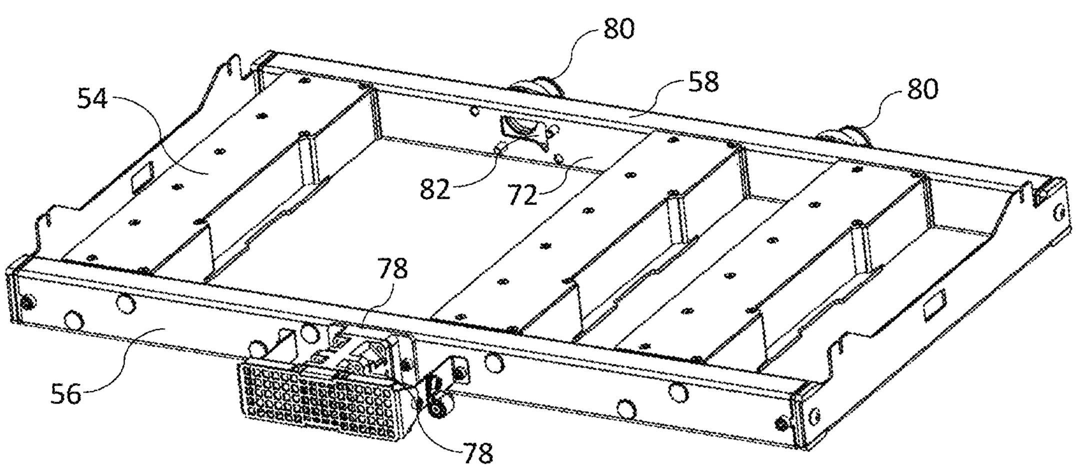


FIG. 12

# PRINTING UNIT WITH COMPACT CABLE MANAGEMENT

## PRIORITY

This application claims the benefit of priority to U.S. Provisional Patent Application Ser. No. 63/512,536 filed Jul. 7, 2023, entitled "Compact Printing Unit with Integrated Aerosol Extraction Unit", the contents of which being incorporated herein by reference in its entirety.

## FIELD OF THE DISCLOSURE

This disclosure relates to a printing unit with integrated print modules, maintenance modules and aerosol extraction unit for a digital inkjet press. It has been developed primarily to provide a full color digital inkjet press having a minimal footprint for facile integration into an existing analog press.

## BACKGROUND OF THE DISCLOSURE

Inkjet printers employing Memjet® technology are commercially available for a number of different printing formats, including desktop printers, digital inkjet presses and wideformat printers. Memjet® printers typically comprise one or more stationary inkjet printhead cartridges, which are user-replaceable. For example, a desktop label printer comprises a single user-replaceable multi-colored printhead cartridge, a high-speed label printer comprises a plurality of user-replaceable monochrome printhead cartridges aligned along a media feed direction, and a wideformat printer comprises a plurality of user-replaceable printhead cartridges in a staggered overlapping arrangement so as to span across a wideformat pagewidth.

U.S. Pat. No. 10,076,917, the contents of which are incorporated herein by reference, describes a commercial pagewide printing system comprising an N×M two-dimensional array of print modules and corresponding maintenance modules. Providing OEM customers with the flexibility to select the dimensions and number of printheads in an N×M array in a modular, cost-effective kit form enables access to a wider range of commercial digital printing markets that are traditionally served by analog (offset) printing systems.

Nevertheless, it is still desirable to simplify integration of modules into a compact 'drop-in' printing unit. Compact printing units enable them to replace analog printing stations having a similar footprint in existing web feed systems and, moreover reduce development time and costs for OEMs.

Aerosol extraction is a critical system requirement for high-speed inkjet printing systems, especially thermal inkjet systems having relatively small drop sizes (e.g. less than 5 pL). Ink mist generated during printing must be removed as efficiently as possible in order to avoid loss of print quality and potential contamination of downstream print modules. U.S. Provisional Application No. 63/385,854 filed Dec. 2, 2022 (the contents of which are incorporated herein by reference) describes a novel aerosol extractor, which makes use of an air knife to disrupt aerosol trapped in a boundary layer associated with moving media and remove the disrupted aerosol via suction.

It would be desirable to provide a compact full-color inkjet printing unit, which includes an efficient aerosol extraction unit whilst maintaining a minimal overall footprint of the printing unit. Providing a minimal footprint requires compact integration not only of mechanical com-



ponents of the printing unit, but also electrical cabling and ink lines supplying power, data and ink to printheads.

## SUMMARY

In one aspect, a printing unit is disclosed. In one embodiment, the printing unit includes:

- a fixed maintenance chassis having a bottom plate defining a plurality of first slots;
- a plurality of print modules, each print module having a respective printhead extending across a media feed path;
- a lift mechanism for lifting and lowering the print modules relative to the media feed path; and
- an aerosol extraction unit mounted to an underside of the bottom plate, the aerosol extraction unit including a plurality of aerosol extractors extending across the media feed path between a pair of side bars, the plurality of aerosol extractors being spaced apart along the media feed direction to define a corresponding plurality of second slots aligned with the first slots,
- wherein each print module is slidably movable through its respective first and second slots towards the media feed path into a printing position.

The printing unit according to one aspect advantageously enables a compact configuration for multiple print modules with efficient aerosol extraction for each print module whilst having a minimal overall footprint of the printing unit.

In another aspect, an aerosol extraction unit for an inkjet printer having one or more printheads is disclosed. In one embodiment, the aerosol extraction unit includes:

- a frame having first and second opposite side bars extending parallel with a media feed direction; and
- one or more aerosol extractors extending perpendicular to the media feed direction between the first and second side bars, each aerosol extractor including:
  - an air knife having a knife slot for directing a flow of air towards print media; and
  - a suction nozzle positioned upstream of the knife slot for directing aerosol into a suction channel,

wherein:

- the first side bar comprises an air manifold in fluid communication with each knife slot; and
- the second side bar includes a suction manifold in fluid communication with each suction nozzle.

The aerosol extraction unit according to one aspect provides a compact unit for multiple printheads, which leverages the novel aerosol extractor described in U.S. Provisional Application No. 63/385,854 filed Dec. 2, 2022 (the contents of which are incorporated herein by reference).

In a third aspect, a printing unit includes:

- a fixed unit chassis having a lower cable tray mounted to a front part thereof;
- a movable printhead chassis having an upper cable tray mounted to a front face thereof, the printhead chassis supporting an upstream print module and a downstream print module relative to a media feed direction;
- a lift mechanism for lifting and lowering the printhead chassis relative to the unit chassis;
- upstream and downstream cable trunkings positioned at opposite upstream and downstream ends of the unit chassis for receiving electrical cabling from the lower cable tray; and
- upstream and downstream conduits connected to the upstream and downstream print modules respectively,
- wherein the downstream cable trunking feeds electrical cabling from the lower cable tray into the upstream

conduit via the upper cable tray, and the upstream cable trunking feeds electrical cabling from the lower cable tray into the downstream conduit via the upper cable tray.

The printing unit according to one aspect advantageously provides efficient and compact management of electrical cabling in a system having multiple print modules supported on a movable printhead chassis.

As used herein, the term "ink" is taken to mean any printing fluid, which may be printed from an inkjet printhead. The ink may or may not contain a colorant. Accordingly, the term "ink" may include conventional dye-based or pigment-based inks, infrared inks, fixatives (e.g. pre-coats and finishers), 3D printing fluids, biological fluids and the like.

As used herein, the term "mounted" includes both direct mounting and indirect mounting via an intervening part.

## BRIEF DESCRIPTION OF THE DRAWINGS

One embodiment of the present disclosure will now be described by way of example only with reference to the accompanying drawings, in which:

FIG. **1** is a top perspective view of a printing unit according to one embodiment in its printing position;

FIG. **2** is a bottom perspective of the printing unit in its printing position;

FIG. **3** is a top perspective of the printing unit in its maintenance position;

FIG. **4** is a bottom perspective of the printing unit in its maintenance position;

FIG. **5** is a magnified top perspective of the printing unit showing the bottom cable tray with cover removed;

FIG. **6** is a magnified top perspective of the printing unit with the bottom cable tray removed to reveal the maintenance chassis;

FIG. **7** is a bottom perspective of the printing unit with the extraction unit and bottom plate removed;

FIG. **8** is a bottom perspective of the printing unit with only the extraction unit removed;

FIG. **9** is a front perspective of the aerosol extraction unit in isolation;

FIG. **10** is a front sectional perspective of part of the aerosol extraction unit;

FIG. **11** is a rear perspective of the aerosol extraction unit with one of the aerosol extractors removed; and

FIG. **12** is front perspective of the aerosol extraction unit with one of the aerosol extractors removed.

## DETAILED DESCRIPTION

### Inkjet Printing Unit 1

Referring to FIGS. **1** to **4**, there is shown an inkjet printing unit **1** according to one aspect of the present disclosure. The printing unit **1** comprises a unit chassis **3** having a pair of rigid upstream and downstream support arms **5**A and **5**B extending away from a back plate **7** perpendicular to a media feed direction indicated by arrow M. A maintenance chassis **9** is mounted between the support arms **5**A and **5**B so as to extend along a length of the printing unit **1**. The maintenance chassis **9** supports four maintenance modules **11**, typically of the type described in U.S. Pat. No. 10,076,917, the contents of which are incorporated herein by reference. Referring briefly to FIGS. **6** to **8**, each maintenance module **11** has an L-shaped base frame **13** fixed to a bottom plate **10** of the maintenance chassis **9**, with each maintenance module supporting a respective capper **15** and wiper **17**.

Returning to FIGS. **1** to **4**, a printhead chassis **20** is liftably mounted to the back plate **7** via a lift mechanism. The lift mechanism comprises a lift motor **22** operatively connected to a lead screw mechanism **24**, which reciprocally lifts and lowers the printhead chassis **20** along a nominal z-axis between a printing position (FIGS. **3** and **4**) and a maintenance position (FIGS. **1** and **2**). Mounting blocks **26** of the printhead chassis **20** slidably engage with a pair of parallel guide rails **28** fixed to the back plate **7** of the unit chassis **3** for accurate and repeatable positioning of the printhead chassis **20** in its printing position.

The printhead chassis **20** supports four print modules **30** (CMYK), which are mounted to the printhead chassis vis respective fixed sleeve mounts **31**. Typically, each print module **30** is of the type described in U.S. Pat. No. 10,076,917 and comprises a respective monochrome pagewide inkjet printhead **32** configured to extend across a media feed path as well as a supply module for supplying ink and power/data to each printhead. The print modules **30** are slidably removable from their respective fixed sleeve mounts **31** to allow printhead removal and replacement, as required.

Each print module **30** receives power, data and ink via a respective conduit **40**, which houses electrical cabling and ink lines (not shown for clarity). Each conduit **40** is detachably connected to its respective print module **30**, thereby enabling convenient replacement of printheads **32** via detachment of the conduit and removal of the print module from its respective sleeve mount **31**. An upstream pair of conduits **40**A bend downwards towards an upper cable tray **42** mounted to a front part of the printhead chassis **30**, while a downstream pair of conduits **40**B bend downwards in an opposite direction towards the upper cable tray. A downstream cable trunking **44**B (e.g. "energy chain" or "cable drag chain") is fixed to the downstream support arm **5**B and feeds electrical cabling and/or ink lines (not shown) to the upstream print modules via respective upstream conduits **40**A, while an upstream cable trunking **44**A is fixed to the upstream support arm **5**A and feeds electrical cabling and/or ink lines to the two downstream print modules via respective downstream conduits **40**B. The upstream and downstream cable trunkings **44**A and **44**B receive electrical cabling and/or ink lines from a lower cable tray **46** fixed to a front part of the support arms **5**A and **5**B, thereby providing compact management of electrical cabling and/or ink lines, whilst allowing movement of the printhead chassis **20** along the z-axis for printhead maintenance. A front face of the fixed lower cable tray **46** has a power connector assembly **47**, data connectors **48** and an ink connector assembly **49** for convenient connection of power cables, data cables and ink lines, respectively, via a user-accessible front face of the printing unit **1**.

As best shown in FIG. **8**, the maintenance chassis **9** includes the bottom plate **10**, which defines four first slots **12** corresponding to each of the four printheads **32**. An aerosol extraction unit **50** is fixedly mounted to an underside of the bottom plate **10**, as shown in FIGS. **2** and **4**. The aerosol extraction unit **50** comprises four aerosol extractors **54** extending parallel with each of the four printheads **32** and spaced apart along the media feed direction M to define four second slots **52** aligned with the corresponding first slots **12**. Each print module **30** is slidably movable, by means of the lift mechanism engaged with the printhead chassis **20**, through its respective first and second slots **12** and **52** towards a media feed path into a printing position, as shown in FIGS. **1** and **2**. With the print modules **30** raised above the first and second slots **12** and **52**, as shown in FIGS. **3** and **4**, the printing unit **1** is configured in its maintenance position for capping and/or wiping the printheads **32**.

Aerosol Extraction Unit **50**

The aerosol extraction unit, shown in isolation in FIG. **9**, comprises a frame having first and second (front and rear) side bars **56** and **58** extending parallel with the media feed direction M and a pair of end plates **59** interconnecting the side bars. Four aerosol extractors **54** corresponding to the four print modules **30** extend perpendicular to the media feed direction between the side bars **56** and **58**, each aerosol extractor being at least coextensive with its corresponding printhead **32**. Each aerosol extractor **54** is generally of the type described in U.S. Provisional Application No. 63/385,854 filed Dec. 2, 2022, the contents of which are incorporated herein by reference. Accordingly, and referring now to FIG. **10**, each aerosol extractor **54** includes an air knife having a knife slot **60** for directing a flow of air towards print media and a suction nozzle **62** positioned upstream of the knife slot for directing aerosol into a suction channel **64**. As described in U.S. Provisional Application No. 63/385,854, the air knife serves to disrupt ink aerosol trapped in a boundary layer associated with fast-moving print media by providing a relatively high velocity airflow directed onto the media, while the suction nozzle **62** extracts the disrupted aerosol from a region upstream of the air knife using a relatively lower velocity airflow. As shown in FIG. **10**, each suction nozzle **62** is positioned higher than its respective knife slot **60** relative to the media feed path and a convex guide surface **66** extends between an upstream side of each knife slot and a downstream side of each suction nozzle. The disrupted aerosol tends to follow the guide surface **66** into the suction nozzle **62** by virtue of the Coanda effect.

For compact integration of multiple aerosol extractors, and referring now to FIGS. **11** and **12**, the first side bar **56** comprises an air manifold **70** in fluid communication with each one of the knife slots **60**, while the second side bar **58** comprises a suction manifold **72** in fluid communication with each one of the suction nozzles **62**. The air manifold **70** has an air inlet port **74** for connection to a pressurized air source and four air outlets **76** connected to four corresponding air channels **77** of the aerosol extractors **54**. The air inlet port **74** incorporates a blower fan and filter assembly **78** for delivering clean, pressurized air into the air manifold **70**. The suction manifold **72** comprises a pair of suction ports **80** for connection to a vacuum source (not shown) and suction openings **82**, which provide suction to respective suction channels **64** of the four aerosol extractors **54**. The suction ports **80** may be connected to the vacuum source via suitable tubing (not shown) extending through an opening in the back plate **7** of the printing unit **1**.

As shown in FIG. **10**, each aerosol extractor **54** comprises an extractor housing **84** having a suction chamber **86** connected to one of the suction openings **82** of the suction manifold **72**. The suction channel **64** is connected to the suction chamber **86** and thence to the suction manifold **72** via the suction openings **82**. Gutters **88** extend along a length of each suction chamber **86** for collecting aerosol deposited on internal surfaces thereof. In addition, an elongate baffle plate **90** extends along a length of the suction chamber **86** in order to provide a relatively even suction force along its length. In use, the aerosol extraction unit **50** provides compact and efficient aerosol extraction for each printhead **32**, whilst allowing the printheads to extend into the printing position and retract into the maintenance position.

From the foregoing, it will be appreciated that the printing unit **1** provides a highly compact design with close positioning of printheads along the media direction and a minimal footprint suitable for incorporation into existing web feed systems.

It will, of course, be appreciated that the present disclosure has been described by way of example only and that modifications of detail may be made within the scope of the present disclosure, which is defined in the accompanying claims.

The invention claimed is:

**1**. A printing unit comprising:

a fixed unit chassis having a lower cable tray mounted to a front part thereof;

a movable printhead chassis having an upper cable tray mounted to a front face thereof, the printhead chassis supporting an upstream print module and a downstream print module relative to a media feed direction;

a lift mechanism for lifting and lowering the printhead chassis relative to the unit chassis;

upstream and downstream cable trunkings positioned at opposite upstream and downstream ends of the unit chassis for receiving electrical cabling from the lower cable tray; and

upstream and downstream conduits connected to the upstream and downstream print modules respectively,

wherein the downstream cable trunking feeds electrical cabling from the lower cable tray into the upstream conduit via the upper cable tray, and the upstream cable trunking feeds electrical cabling from the lower cable tray into the downstream conduit via the upper cable tray.

**2**. The printing unit of claim **1** comprising a pair of upstream print modules and a pair of downstream print modules, each print module having a respective conduit for feeding electrical cabling thereto.

**3**. The printing unit of claim **1**, wherein each conduit feeds one or more ink lines together with the electrical cabling to its respective print module.

**4**. The printing unit of claim **1**, wherein each conduit is detachably connected to its respective print module.

**5**. The printing unit of claim **1**, wherein the lower cable tray comprises a front face having at least one of: a power connector, a data connector and an ink connector.

**6**. The printing unit of claim **1**, wherein the first and second cable trunkings bend downwards towards the lower cable tray, each cable trunking having an axis of curvature extending parallel with a media feed direction.

* * * * *